Simulation Study Plan

shg_mode_survey · geometry × material design matrix, agreed 2026-10-12

This documents the campaign -- originally ten geometry/material combinations, now 13 after splitting row 5 into 3 Sc-composition levels and adding row 1.2 (bare rib, 2026-10-14) (MATRIX, next page), grouped into four underlying cross-section topologies (one page each, with a real diagram where the family already exists, a schematic where it doesn't yet).

Built from the generalized-geometry sketches shared 2026-10-12: a 3-layer stack (A = base film/rib, C = core ridge, B = loading/shell region), where whether B just caps C's top or wraps around its sides, and how large B is, span both the guided-mode families already in this repo (ridge, loaded, the corrected hybrid) and the not-yet-built Cerenkov case in one consistent vocabulary -- not four unrelated structures.

READING THE MATRIX

status: ✓ ready to run now, with existing materials data.
○ blocked on a materials gap (d33(Sc), GaAlN n/k/d33, or an anisotropic-substrate decision) -- see the Input Assumptions report for what's known and what isn't.
h_A: the base-layer thickness, free 0 (pure ridge/no embedding base) up to a full film (rib/loaded) -- only wired up in the `loaded` family today; adding it to the embedded topologies (rows 6-9) is new geometry work, not yet done (marked TBD*).

STANDING ASSUMPTION FOR EVERY NEW RUN FROM HERE ON

AlN's own measured SH-band absorption (AM187-LT6 k_e) was explored and REJECTED (it implied ~100,000 dB/cm, checked against real device output and found 3-4 orders of magnitude too high). Every row below should instead launch with geom.params['sh_absorption_calibrated'] = True -- a flat, empirically-calibrated 100 dB/cm SH-band addition, confinement-weighted, layered on top of native scattering (unaffected). See Input Assumptions report section 3b and PLAN.md's 2026-10-12 entries for the full derivation.

Not a commitment to run all ten at once -- user direction, 2026-10-12: check in before each batch.

Design matrix

13 rows, grouped by topology — see the following pages for diagrams of each

#	TOPOLOGY	H_A	CORE (C)	SHELL/OUTER (B)	SUBSTRATE	PURPOSE	STATUS
1 ✓	Ridge	n/a (no A layer)	AlN	—	Al2O3	baseline	done (`row1_ridge`); loss comparison done
1.2 ✓	Ridge	0 → film thickness	AlN	— (no strip, B=0)	Al2O3	bare partial-etch rib, no strip -- between row 1 and row 4	done; best NCE 2.25, peak_eff up to 0.0038 %/W (~row-1-like, no strip = no bulk loss)
2 ✓	Ridge	n/a	ScAlN	—	Al2O3	Sc-enhanced d33 alone, no loading	ready — d33(Sc) set, AT RISK
3 ✓	Rib-loaded	0 → film thickness	AlN	TiO2	Al2O3	baseline	done — found bad (SH-absorbing)
4 ✓	Rib-loaded	0 → film thickness	AlN	SiN	Al2O3	SiN strip instead of TiO2	done; best NCE 9.49 %/W/cm², Cerenkov-flagged (structural)
5.1 ✓	Rib-loaded	0 → film thickness	AlN	ScAlN (9% Sc)	Al2O3	LOW: index boost to pull the mode up	done; best NCE 1.57 %/W/cm² (below row 4)

Design matrix (cont. 2)

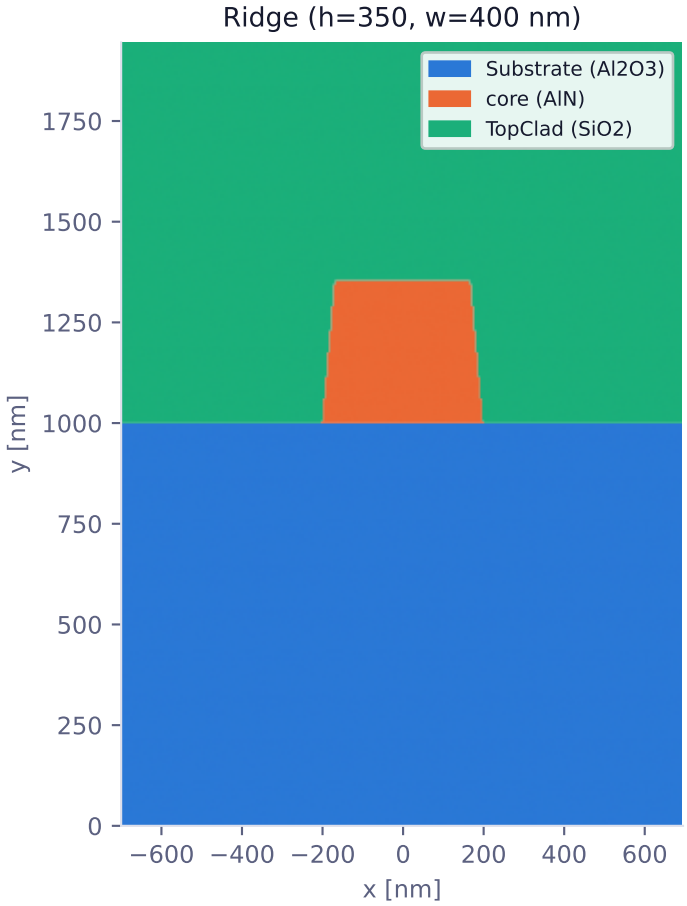
#	TOPOLOGY	H_A	CORE (C)	SHELL/OUTER (B)	SUBSTRATE	PURPOSE	STATUS
5.2✓	Rib-loaded	0 → film thickness	AlN	ScAlN (41% Sc)	Al2O3	HIGH: chase the d33 boost	done; best NCE 2313 %/W/cm² (~244x row 4), d33 EXTRAPOLATED
5.3✓	Rib-loaded	0 → film thickness	AlN	ScAlN (23% Sc)	Al2O3	MIDDLE: fill the low/high gap	done; best NCE 11.2 %/W/cm² (above row 4, nowhere near 5.2 — grid-specific overlap, not monotonic with Sc%)
6 ✓	Embedded (guided)	0 → TBD*	AlN	SiN	Al2O3	corrected `hybrid` — Honda et al. mechanism	done (`row6_hybrid_lateral`/`_vertical`); TM20 not found, close analogues
7 ✓	Embedded (guided)	0 → TBD*	AlN	ScAlN	Al2O3	both-active variant	ready — d33(Sc) set, AT RISK
8 ○	Embedded (Cerenkov)	0 → TBD*	ScAlN	AlN (blanket)	Al2O3 or AlN	buried high-index channel radiating into thick AlN	d33 set; needs embedded-Cerenkov geometry (not built)

Design matrix (cont. 3)

#	TOPOLOGY	H_A	CORE (C)	SHELL/OUTER (B)	SUBSTRATE	PURPOSE	STATUS
9 ○	Embedded (Cerenkov)	0 → TBD*	GaAlN	AlN (blanket)	Al2O3 or AlN	same, GaAlN channel	need full GaAlN n/k/d33
10 ○	any (cross-cutting)	—	—	—	AlN instead of Al2O3	substrate swap	eqns exist, anisotropic treatment TBD

Topology 1: Ridge

real cross-section, from geometry.py



MATRIX ROWS

✓ **#1** AlN / — — baseline

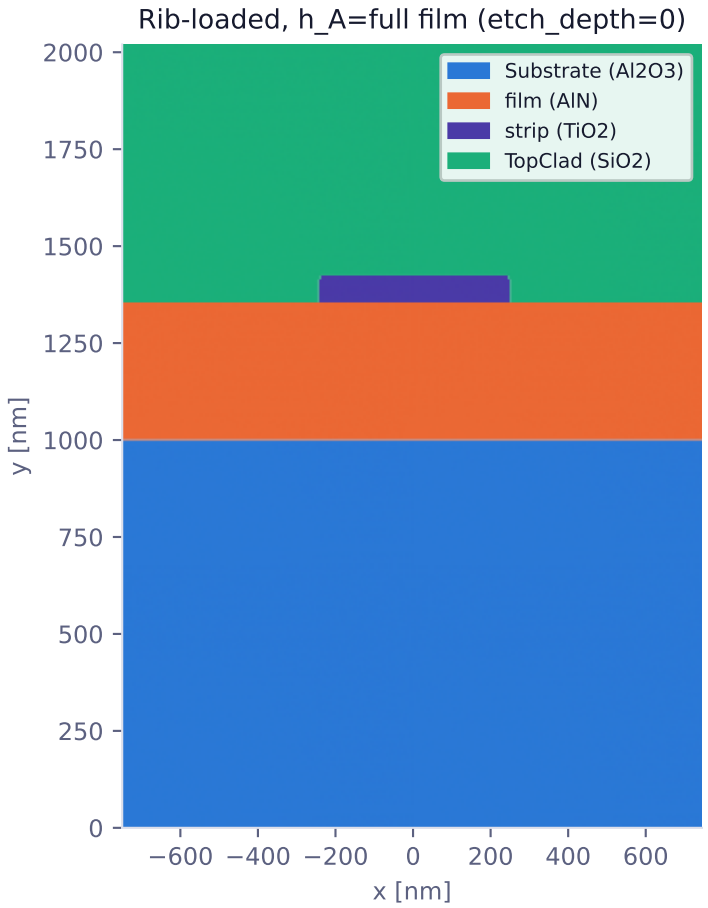
✓ **#1.2** AlN / — (no strip, B=0) — bare partial-etch rib, no strip -- between row 1 and row 4

✓ **#2** ScAlN / — — Sc-enhanced d33 alone, no loading

Bare AlN (or ScAlN) core directly on substrate -- no A layer, no B/shell at all. Still the best-surviving structure under real (calibrated) SH loss (aln_loss_report.pdf): median calibrated peak efficiency ~41,000x above `loaded`'s. Row 2 asks whether Sc-doping the core alone beats even this -- ready, 32%-Sc working d33 (section 6b, AT RISK). Row 1.2 asks a different question: a PARTIALLY etched bare rib (no strip) -- the B=0 edge case of the Rib-loaded topology (row 4), grouped here since it's still bare AlN. Needs a `geometry.Loaded` no-strip code path (not yet done) -- user-approved 2026-10-14, building now. SWEPT RANGE (row 1, done): h_core 300-400nm x w_core 300-800nm -- best h=350/w=500, NCE 14.9 %/W/cm^2 (num_sh_modes 30->45 fix confirmed this unchanged from the v1 data). Row 1.2: t_film range TBD. Row 2: not yet run, row 1's range proposed.

Topology 2: Rib-loaded (cap-only)

real cross-section, from geometry.py



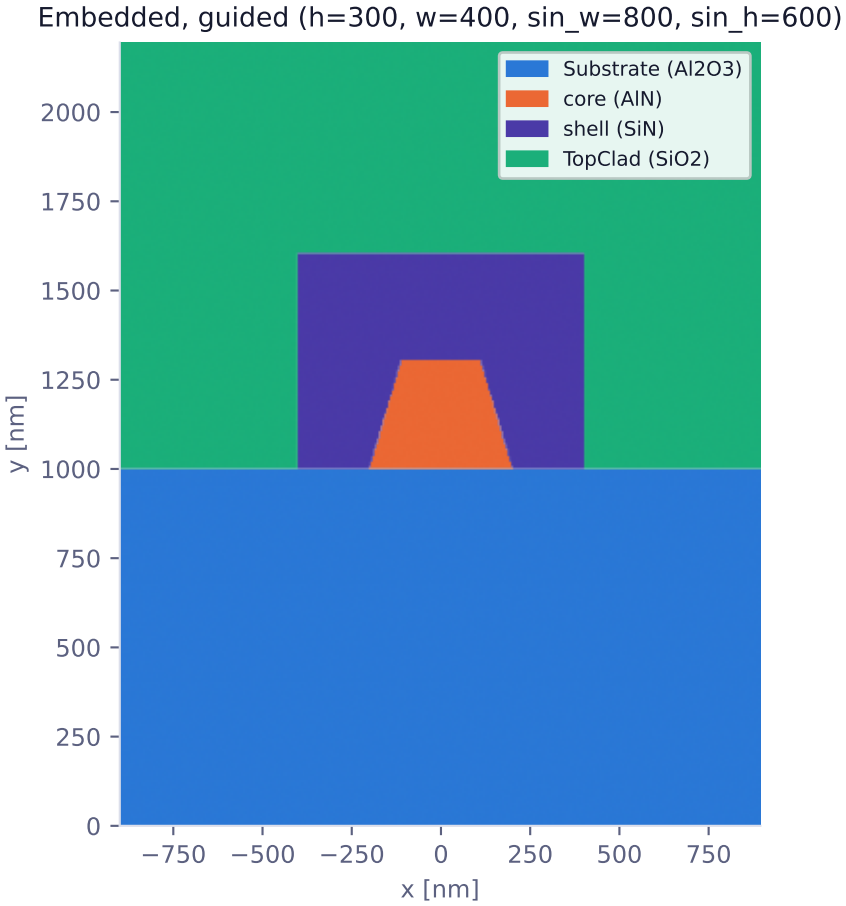
MATRIX ROWS

- ✓ **#3** AlN / TiO2 — baseline
- ✓ **#4** AlN / SiN — SiN strip instead of TiO2
- ✓ **#5.1** AlN / ScAlN (9% Sc) — LOW: index boost to pull the mode up
- ✓ **#5.2** AlN / ScAlN (41% Sc) — HIGH: chase the d33 boost
- ✓ **#5.3** AlN / ScAlN (23% Sc) — MIDDLE: fill the low/high gap

A (AlN film) with a loading strip B capping only its top -- the `loaded` family, always run at etch_depth=0 (unetched film) so far (the B=0/full-etch extreme is row 1.2 instead). Row 3 (TiO2) found bad -- SH-absorbing, ~40,000x below ridge's. Row 4 (SiN) -- done, NCE 9.49, but linearly inert (no chi(2)). Row 5 (ScAlN, nonlinear-active) split into 3 compositions (2026-10-14): 5.1 (9%, done, NCE 1.57, below row 4), 5.2 (41%, done, NCE 2313 ~244x row 4 -- real d33 boost, verified via edge-ratio + effective-area checks, not an artifact; d33 EXTRAPOLATED), 5.3 (23%, next, d33 INTERPOLATED -- lower risk). ALL refined crossings in rows 4/5.1/5.2 are Cerenkov-flagged -- structural to etch_depth=0 (an extended same-material AlN slab always exists), not material-specific; unquantified (no FDFD rate solve run yet).
SWEPT RANGE: t_film {354,436,518,600}nm x h_s {70,150}nm x w_s {500,1000}nm, etch_depth=0 -- identical across rows 3/4/5.1/5.2/5.3 so only the strip material/composition varies.

Topology 3: Embedded (guided)

real cross-section, from geometry.py



MATRIX ROWS

✓ **#6** AlN / SiN — corrected `hybrid` — Honda et al. mechanism

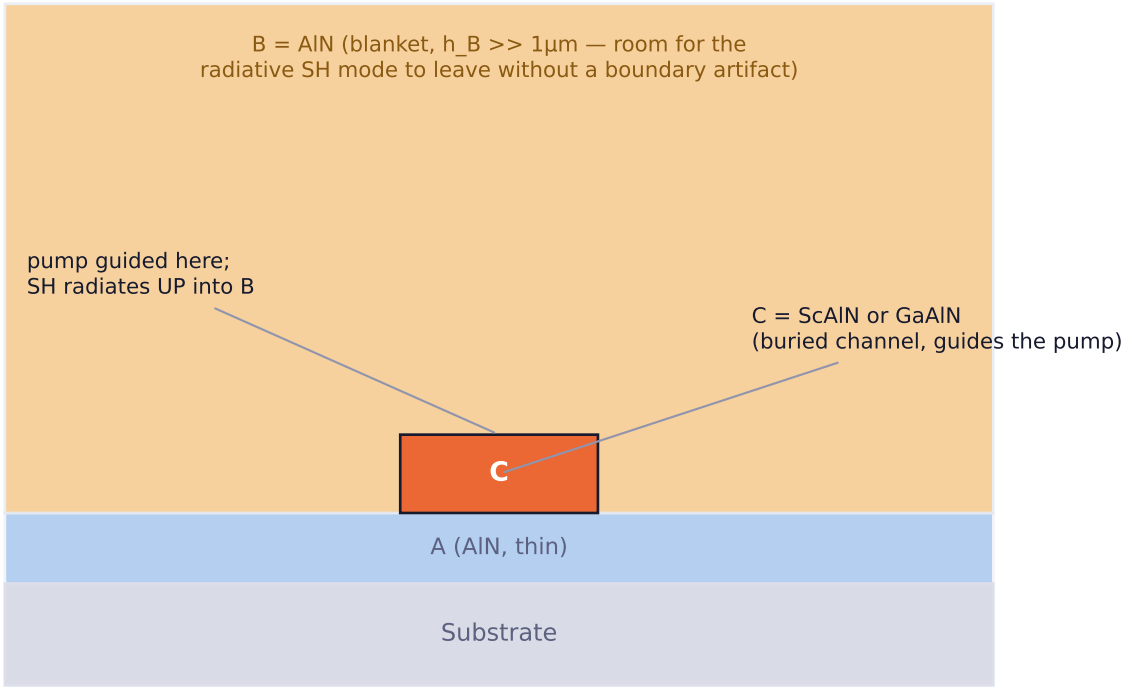
✓ **#7** AlN / ScAlN — both-active variant

Core C wrapped on the sides (and above) by shell B (w_C <= w_B) -- the corrected `hybrid` family, fixed 2026-10-10 after the earlier 'pedestal' geometry bug (core was stacked on top of the shell, not embedded in it -- see PLAN.md). Tests the Honda et al. 2025 TM00->TM20 mechanism; the re-run (row 6) is the next EMode campaign queued, launching with the standing sh_absorption_calibrated=True assumption from the start (already wired into both configs below). Row 7 (ScAlN shell) is ready too, using the 32%-Sc working d33 (section 6b of Input Assumptions, AT RISK).

SWEPT RANGE: lateral (`hybrid_sweep`) w_core 200-600 x sin_width 700-1500nm, fixed h_core=400/sin_height=700, 24 pts; vertical (`hybrid_vertical`) h_core 250-450 x sin_height 450-1100nm at the lateral sweep's best point, 23 pts (450 added to probe the pedestal data's "smaller wins" trend to the 200nm-margin boundary).

Topology 4: Embedded (Cerenkov)

SCHEMATIC — not yet built



MATRIX ROWS

- ○ **#8** ScAlN / AlN (blanket) — buried high-index channel radiating into thick AlN
- ○ **#9** GaAlN / AlN (blanket) — same, GaAlN channel

Same wrap-around embedding as Topology 3, but the material roles flip and B grows large: C (small, buried) carries the enhanced index/nonlinearity (ScAlN or GaAlN) and guides the pump; B becomes a blanket AlN cladding thick enough (>>1 um) to host a genuinely radiative SH continuum rather than a tightly confined guided mode. No new plumbing needed for the Cerenkov physics itself -- `cerenkov_fdfd.py`'s driven 2D solve already takes eps_fn/d_fn/bbox generically from any geometry class, the same interface every existing family implements. What's missing is the geometry class itself (not yet written) and, for row 9 specifically, a live transparency check -- GaAlN's favorable (higher-d33) direction is Ga-rich, but GaN's bandgap sits below the SH photon energy, so the useful window may be narrow or may not exist; not yet checked quantitatively (Input Assumptions report, section 5).

Sequencing

Proposed order, each step its own check-in (user direction, 2026-10-12: not a green light to run all rows unsupervised) -- every step launches with geom.params['sh_absorption_calibrated']=True from the start (page 1), not reprocessed later --

1. DONE (2026-10-12/13): Row 6, `row6_hybrid_lateral`/`row6_hybrid_vertical` -- 47 geometries, 376 refined crossings, zero errors. TM00->TM20 itself not found; closest analogues are TM0x->TM2(5-9). See runs/hybrid_subfigure_report.pdf.
2. DONE (2026-10-13): Row 1 refresh, `row1_ridge` -- num_sh_modes 30->45 fixed a coverage gap found in `row1_ridge_v1`'s log; best crossing unchanged (h=350/w=500, NCE 14.9).
3. DONE (2026-10-13/14): Row 4 (`row4_loaded_sin`, SiN) -- best NCE 9.49. Row 5 split into 3 Sc-composition levels (2026-10-14): 5.1 (9%, low) and 5.2 (41%, high) both done -- best NCE 1.57 and 2313 respectively, the latter a confirmed real d33 boost (~244x row 4, verified against edge-ratio and effective-area artifacts). ALL refined crossings in rows 4/5.1/5.2 Cerenkov-flagged (structural, etch_depth=0 -- see Topology 2 page).
4. NEXT (2026-10-14, user-approved): Row 1.2 (bare partially-etched rib, no strip -- needs a geometry.Loaded code change, building now) and Row 5.3 (23% Sc middle composition, interpolated d33=44.2/d31=2.8 pm/V, lower risk than 5.2's extrapolated value).
5. THEN (2026-10-14, user-approved, geometries TBD -- discuss before launching): Row 8 (embedded-Cerenkov, ScAlN channel) and Row 9 (embedded-Cerenkov, GaAlN channel). Row 8's d33 is set (same AT RISK value); both still need the embedded-Cerenkov geometry class (topology 4) built -- doesn't exist yet. Row 9 also needs GaAlN n/k sourced first (Input Assumptions report, section 5).
6. Row 2/7: ridge and embedded-guided ScAlN variants -- d33 question resolved (section 6b), configs not yet written (mirror row 1/6's grids with ScAlN swapped in). Not yet sequenced relative to rows 8/9 -- revisit after those.
7. Row 10: substrate swap, cross-cutting -- can run alongside any of the above once the anisotropic-vs-isotropic AlN-substrate treatment is settled (user direction, 2026-10-12: anisotropic).